

HWRS 564a Hydrogeologic Analysis Tools & Methods I

Fall 2026

Class time: Tu/Th 2:00 – 3:15

Class Location: Harshbarger 203

Instructor and Contact Information

Dr. Andrew Bennett (he/him)
Office hours: 1:00-2:00 pm, every Tuesday
Office: JW Harshbarger Bldg Rm. 322B
Email: andrbenn@arizona.edu
Website: <https://has.arizona.edu/person/andrew-bennett>

Catalog Description

This course will present foundational tools and methods for analyzing and modeling hydrogeologic systems. Students will learn how to perform quantitative analysis on real world data using Python. Students will also learn to simulate groundwater systems using the USGS MODFLOW hydrologic model. Topics covered will include writing iterative solutions, performing timeseries analysis, plotting figures and graphs, and running model simulations. As a result of these activities students will gain a holistic view of the role of computing and computational methods in understanding hydrogeologic systems.

Course Prerequisites or Co-requisites

No pre-requisites, but the student must be enrolled in the MS Hydrogeology program.

Required co-registration in HWRS 599 Section 001 (Recitation), HWRS 561a - Physical Hydrogeology I, HWRS 562a Chemical Hydrogeology I, HWRS 563a Hydrogeologic Measurement Methods I, and HWRS 565a Communications in Hydrogeology I

Required Textbooks/Materials

None

Reference Readings (Optional)

References will be provided as needed through D2L. You will not need to purchase these references.

Course Objectives

Students will...

1. import data into python, conduct simple calculations and plot data
2. build and run computational notebooks, setting up a python environment and importing the required python packages
3. Use scientific computing packages such as pandas, and matplotlib to analyze and plot timeseries data and gridded hydrogeologic data
4. identify and fill missing or outlier data using python
5. Run and modify steady state and transient MODFLOW models using templates provided by the instructor
6. Understand and use best practices and modern techniques for writing code to conduct scientific analysis

Expected Learning Outcomes

Students will be able to...

1. Manipulate, clean, and analyze major types of hydrogeologic data using common analysis platforms such as python
2. Interpret and modify MODFLOW Models for both transient and steady state systems.
3. write common programming constructs for data analysis and processing such as loops, control flow, aggregation, and filtering.
4. Build data processing pipelines using industry standard techniques that are reproducible, extensible, and documented.

Course Format and Teaching Methods

This course will consist of a mixture of lectures, discussion and interactive coding activities. Most sessions include short coding exercises that you will work through in class, and most weeks begin with a brief quiz on the concepts from the previous session. Every week, I will present new material, students will present their work and we will have group discussions and code walkthroughs. This is a very interactive course and students are expected to participate in discussions and come prepared to share their work and help others.

Planned Field Trips

The class will begin this course with a field trip during the entire first week of class, and will apply to all courses in the MS Hydrogeology Fall semester (this course, but also HWRS 561a, HWRS 562a, HWRS 564a, and HWRS 565a). The field trip will begin on the first Monday of the first full week. Students should plan to be out of town for the entire week, including camping overnight.

Additionally, there may be field trips in the other courses in the MS Hydrogeology program. The timing of trips is coordinated with students' schedules and the instructors of the other courses in the MS Hydrogeology program. Please plan accordingly. Participation in field trips for the other courses will not impact the schedule of this course.

Schedule of Topics & Activities

The course will be organized around month long projects that give a learning context for all five co-convened classes, following the theory-data-prediction structure, including case-based theory and project-based methodology practice. For Fall 2026, the scheduled activities are as follows. Short in-class quizzes are given in weeks 3 through 13 and week 15. Course schedule is subject to change.

Week	Monday date	Topics covered this week	Assignment due this week
1	8/24/2026	Program field trip: no class meetings this week.	
Module 1: Python Foundations			
2	8/31/2026	Getting started with Python: what Python is and how we interact with it, working in GitHub Codespaces and Jupyter notebooks, setting up a Python environment, importing packages, variables, data types, arithmetic, strings and sequences.	
3	9/7/2026	Control flow and functions: using conditional statements to select data, for and while loops, writing and modifying your own functions, accessing mathematical functions from Python packages.	HW 1: My first Jupyter notebook
4	9/14/2026	Arrays and iterative solutions: numpy arrays and array operations, writing an iterative solution (root finding and reservoir recession), when to loop and how to avoid unnecessary loops, precision vs. accuracy.	HW 2: Control flow and functions
5	9/21/2026	Introduction to pandas: reading csv data into Python, pandas indexing conventions and selecting subsets of data, row wise and column wise operations, basics of plotting with matplotlib.	HW 3: Iterative groundwater solution in Python Project 1 Analysis Homework
Module 2: Data Analysis & Visualization			
6	9/28/2026	Working with real world data: retrieving data from web APIs, data wrangling, identifying and filling missing data, detecting and handling outliers.	HW 4: Working with pandas dataframes
7	10/5/2026	Timeseries analysis with pandas: understanding and working with datetime data in Python, working with dates using pandas, groupby, resample and rolling operations, plotting dates.	HW 5: Data retrieval and cleaning
8	10/12/2026	More on matplotlib: adding and adjusting titles, controlling color, adding legends, plotting more than one thing, creating multi panel plots, mapping with cartopy, finding example plots to start from.	HW 6: Pandas groupby, resample, and rolling operations Project 2 Analysis Homework Part 1
9	10/19/2026	Statistics and regression: descriptive statistics, fitting and evaluating a linear regression with scikit-learn, interpreting a fitted model, visualizing water chemistry data.	HW 7: Plotting with matplotlib
Module 3: Introduction to MODFLOW & FloPy			
10	10/26/2026	Intro to FloPy and MODFLOW: basics of the MODFLOW model, what it does and why we use it, overview of MODFLOW's modular structure, ways of interacting with MODFLOW, intro to FloPy, installing and running a MODFLOW model with FloPy.	HW 8: Regression analysis Project 2 Analysis Homework Part 2
11	11/2/2026	Inputs, outputs, and hitting go: using a 2D steady state model we will walk through the basics of setting up, running, and visualizing results from a MODFLOW model.	HW 9: MODFLOW structure diagram

12	11/9/2026	Discretization, initial conditions, and boundary conditions: setting up a simple grid, types of boundary conditions and how you set them in MODFLOW, interpreting and plotting outputs, confined vs. unconfined simulation.	HW 10: My first MODFLOW model
13	11/16/2026	ET, recharge, and pumping in MODFLOW: running transient models, implementing ET and recharge, choosing between MODFLOW packages and understanding their assumptions, inserting recharge and pumping wells, plotting to compare outputs from multiple simulations.	HW 11: Building a 2D steady state MODFLOW model Project 3 Analysis Homework Part 1
14	11/23/2026	Layered and river models: multi-layer simulations, adding a river boundary condition, loading and modifying an existing MODFLOW model. No class Thursday 11/26 — Thanksgiving recess.	
Module 4: Reproducible Workflows			
15	11/30/2026	Debugging and coding practices: reading tracebacks and strategies for debugging, writing clean code, writing your own functions and modules, setting up environments others can use, using Python scripts rather than Jupyter notebooks.	HW 12: Applying fluxes to a 2D MODFLOW model Project 3 Analysis Homework Part 2
16	12/7/2026	Project 3 presentations: mini-conference presentations of the term project. No class Thursday 12/10.	

Course Assessments and Grading Breakdown

You will be assessed on the basis of weekly assignments. You will also be assessed based on how you apply the understanding gained in this class to the projects. Finally, you will also be assessed based on your participation in the course.

Course Assignments: are designed to develop your understanding of elements of the course and to give you practice in applying that knowledge. A time budget will be given for each Course Assignment. They may include calculations, analysis, synthesis, and written elements. Course Assignments count toward your grade in the course and should be turned in through the D2L site for this course unless otherwise indicated by the instructor. The instructor may provide specific format requirements for assignments. In this course, Course Assignments will be given on Thursdays. Assignments that are to be turned in online through D2L will be due at Wednesday at 11:59pm either one or two weeks from the assigned date. Due dates will be clearly posted on D2L.

If you know that you will miss the due date for an assignment, you must contact the instructor at least 12 hours in advance of the due date/time and discuss whether you can turn it in for partial credit. You and the instructor will agree upon the possible credit and the revised due date. When calculating the grades for this course, the instructor will not include the lowest graded assignment. But, that assignment must have been turned in by the due date or by the revised date allowed by an instructor with prior approval. The work must represent an honest effort to complete the assignment. Every assignment is a learning opportunity, so you cannot use a 'drop' to not complete an assignment. Grades cannot be dropped for projects or presentations.

Project-Relevant Assignments: are designed to help you to apply the content of this class to the program-wide projects. In some cases, these Project-Relevant Assignments will produce outcomes that can be included directly in your project. In other cases, the Project-Relevant Assignments will act as a bridge between the course material and the project requirements. Even if the project is team-based, **every student must complete all Project-Relevant Assignments**. A time budget will be given for each Project-Relevant Assignments. They may include calculations, analysis, synthesis, and written elements. Project-Relevant Assignments count toward your grade in the course and should be turned in through the D2L site unless otherwise indicated by the instructor. The instructor may provide specific format requirements for assignments. In this course, Project-Relevant Assignments will be given on Monday. Assignments that are to be turned in online are to be submitted, through D2L, by Wednesday at 11:59pm the following week. Assignments that are to be turned in as hardcopy are due at the beginning of class on the due date.

In-Class Quizzes: are short, low-stakes checks on the concepts covered in the previous session. Each quiz consists of three to five questions, takes about five minutes, and is completed in class through D2L. Quizzes are designed to tell both you and me which ideas have landed and which need another pass, so they are graded on correctness but weighted lightly. Ten quizzes will be given over the semester and your two lowest quiz grades will be dropped. No time budget is given for quizzes; preparing for them is part of your weekly class preparation time.

Participation: it is expected that all students in the program will participate fully in all aspects of the course. This includes showing up in class, being present and engaged in discussion, answering and asking questions during class, and contributing to the culture of learning of the program. If a student is not meeting expectations, they will be notified by the instructor, given guidance on how to increase their participation, and given a chance to improve. Thereafter, if student continues to fall below meeting the expectations for participation, their class participation points will be reduced.

The percentage distribution of your grade will be as follows.

Course Assignments: 60%

Projects – analysis component: 15%

In-Class Quizzes: 10%

Participation: 15%

University policy regarding grades and grading systems is available [at this link](#).

Other Activities – Not Graded/Assessed

Practice Exercises: are designed to help you develop the ability to apply concepts and calculations presented in class. These exercises are NOT GRADED – the answers are provided with the questions. In this course, practice exercises are embedded directly in the lecture slides as runnable code cells, with hints and worked solutions available. No time budget is given for these assignments. Rather, they are provided as an aide in case you feel that you need help. You may be directed to complete these exercises based on the instructor's assessment.

Final Examination or Project

There is no final examination in this course. However, students will be completing 3 projects this semester that cut across all courses in the MS Hydrogeology program, and will require students to utilize and synthesize the skills they learned in all 5 courses to address a hydrogeologic question/problem. Presentation of the third and last Term Project of the semester will take place on December 8, 2026, during the last week of classes. These presentations will be organized as a mini-conference and professional hydrogeologists will be invited to attend in person or online.

Grading Scale

Your final grade will be informed via D2L. Letter grades are determined using the following scale:

A:	>= 90.0%
B:	>= 80 - 89%
C:	>= 70 – 79 %
D:	>= 60 to 69 %
E:	below 59 %

University policy regarding grades and grading systems is available at <https://catalog.arizona.edu/policy/courses-credit/grading/grading-system>.

Latework Policy

This class and the entire program depend strongly on student participation and you are only able to participate fully if you have done the homework on time. Therefore, no late assignments will be accepted for credit. We do understand that life happens, so we will automatically drop your lowest assignment grade for this course when calculating your final grade.

Incomplete (I) or Withdrawal (W):

Requests for incomplete (I) or withdrawal (W) must be made in accordance with University policies, which are available at [this link associated with the registrar](#).

University of Arizona Course Policies

All University of Arizona course and syllabi policies, as well as other helpful information and resources, can be found at [this link](#).

If you are in need of basic needs care, here is [another helpful link](#), in addition to what you can find at the policy link above.

Subject to Change Statement

Information contained in the course syllabus, other than the grade and absence policy, may be subject to change with advance notice, as deemed appropriate by the instructor.